# Unit 1 Review · Kinematics

**Monday, November 2 · Continue unfinished work Tuesday, November 3 · Unit 1 exam Thursday, November 5**

Name: _____ Section: _____

> **Core for both sections:** complete 1–6. Honors: also complete H1–H2. Use separate paper if needed. Write **K**nowns with units, **D**on't know, **E**quation in symbols, **A**lgebra then substitution. Choose positive directions. Unless stated otherwise, ignore air resistance and use $a_y = -9.8$ m/s² when up is positive.

## Monday · Diagnose and practice (44 minutes)

0–5 min: label your confidence on graphs / acceleration / free fall / projectiles. 5–18: problems 1–2. 18–35: problems 3–4. 35–41: compare methods and correct one error. 41–44: write your priority question for Tuesday. Problems 5–6 continue Tuesday.

### 1. Position, distance, and displacement

A walker moves along a straight path. Their position is 0 m at 0 s, 12 m at 4 s, 12 m at 6 s, and 4 m at 10 s. Connect the points with straight segments on a position–time graph. Find the velocity on each segment, total distance, displacement, average speed, and average velocity. Label graph axes and units.

---

### 2. Read the velocity–time model

A cart's velocity rises linearly from 0 to +6 m/s during 0–3 s, stays +6 m/s during 3–5 s, then falls linearly to −2 m/s during 5–9 s. Sketch the graph. Find acceleration on each interval; the time when the cart reverses direction; signed displacement and total distance over 0–9 s. Show the areas.

---

# Acceleration and free fall

## 3. Choose the equation

A cart traveling at +2.0 m/s accelerates uniformly at +1.5 m/s² for 4.0 s. Find its final velocity and displacement. Then explain why final velocity multiplied by time does not give the displacement. Use KDEA.

---

## 4. At the top

A ball is thrown straight upward at 14.7 m/s from a balcony 1.20 m above the ground. Find the time to the highest point, maximum height above the ground, and speed and direction just before it reaches the ground. At the highest point, state both velocity and acceleration. Use KDEA.

---

# Tuesday · Projectiles and evidence

## 5. Two axes, one clock

A ball leaves a level table horizontally at 4.0 m/s. The tabletop is 1.25 m above the floor. Find the flight time, horizontal distance to impact, and vertical velocity at impact. Draw separate x and y knowns. Explain why horizontal speed does not determine flight time.

---

## 6. Use the cart lab

A velocity–time best-fit line for a cart is $v = (0.80\ \text{m/s}^2)t + 0.20\ \text{m/s}$. Identify the acceleration and initial velocity, predict velocity at 2.0 s, and calculate displacement from 0 to 2.0 s. Name one measurement limitation and explain how it could affect the result.

---

## Honors extensions

Complete after the common core. Show the derivation, units, and physical interpretation.

### H1. Launch at an angle

A ball is launched at 20.0 m/s at 30.0° above horizontal and lands at its launch height. Resolve the initial velocity, derive the nonzero flight time from the vertical equation, and find range and maximum height. Explain why the same-height range formula cannot be used unchanged for a landing point 3.0 m lower.

---

### H2. Chase model

Runner A passes x = 0 at constant +4.0 m/s at t = 0. Runner B starts at x = 0 from rest at the same instant and accelerates at +1.0 m/s². Write both position functions; find when B catches A after the start, the meeting position, and B's velocity then. Interpret the t = 0 solution.

---

Exit reflection: One error I corrected: _____
One question I still have: _____
My first study target: _____